Laura Best

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OBJECTIVE

Motivated Electrical Engineering student specializing in mechatronics, PCB design, and embedded systems, seeking opportunities in tech where I can contribute to hardware/software development and real-world engineering solutions. Authorized to work in both Canada and the United States.

EDUCATION

Bachelor of Science, Electrical Engineering
Minor in Mechatronics — Expected graduation: April 2030
Schulich School of Engineering, University of Calgary — Cumulative GPA: 3.9

SKILLS

- Programming Languages: Python, JavaScript, C, C++
- Tools and Environments: Git, GitHub, Visual Studio, Linux/Unix Shell
- Other: Microsoft Office, SOLIDWORKS, Fusion 360, MATLAB, Altium
- Transferable: Documentation, Revision Control, Teamwork, Communication, Time Management, Adaptability

TECHNICAL EXPERIENCE

Telemetry/Cooling Sub Team
ZEUS Electric Motorsport Club, Calgary, Alberta — September 2025 – Present

- Developed and maintained technical documentation for multiple PCBs, including schematics, revision tracking, and manufacturing handoff packages.
- Designed and reviewed telemetry and cooling-system PCB components using Altium Designer to support reliable data acquisition and system monitoring.
- Contributed to cooling-system mechanical and electrical component design, supporting thermal management and subsystem integration.
- Collaborated with electrical team members to align sensor data, CAN signals, and board-level designs with telemetry requirements.

ACADEMIC PROJECTS

Multi-File Data Analytics Interface (Python, NumPy, Matplotlib)

- Engineered a Python application to process and cross-analyze multiple large CSV datasets using NumPy for high-performance numerical computation.
- Designed a custom user interface enabling dynamic comparison of metrics across datasets, including automated calculation of averages and other statistical measures.
- Integrated Matplotlib to generate automated, user-driven visualizations, including comparative graphs.

Remote Controlled Boat (CAD, SOLDERING, Raspberry Pi Pico)

- Developed and integrated onboard electronics using a Raspberry Pi Pico, including motor control, power management, and receiver interfaces.
- Designed organized power and signal wiring to reduce electrical noise and improve serviceability.
- Engineered the hull structure in CAD to ensure proper fit, mounting points, and protection for electronics.
- Integrated joystick inputs with receiver and control logic to achieve smooth and responsive maneuvering.

Fitness Tracking Watch (ESP32-S3, C++)

- Designed and programmed a custom fitness watch using an ESP32 microcontroller and C++ with real-time step tracking based on accelerometer data.
- Built a responsive touchscreen interface for displaying activity metrics and interacting with device features.
- Implemented Bluetooth connectivity for syncing data and enabling communication with external devices.

Campus Events Web App (JavaScript, React, Firebase)

- Built a campus events web application using JavaScript, React, Firebase Firestore, and Firebase Authentication with real-time event updates.
- Implemented core features for creating events, adding friends, and marking interest levels (going, thinking about going, not interested).
- Developed an Explore page and a Friends Activity page to display events and show how friends have interacted with them.
- Designed a clean, responsive UI using React Router and modular components supported by a simple, scalable Firestore data structure